



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	142	Media
HackerNews	11	Community
The Verge AI	6	Media
TechCrunch AI	4	Media
MIT Tech Review	2	Media

VERIFIED INTELLIGENCE — 5 stories

VERI

Hugging Face is being used to easily undress women and children

The Verge AI +2 sources

100%

Hugging Face is being used to make nonconsensual deepfakes, and the popular open-source AI model repository is doing very little to prevent it. That's according to a new report published by the European nonprofit AI Forensics, which found that seven out of the top nine image...

<https://www.theverge.com/ai-artificial-intelligence/971723/hugging-face-nudify-deepfake...>

VERI

Show HN: Segue - Save context in one AI, load it in another by a short handle

HackerNews +1 source

90%

HN 30

Hi HN, MustUse.ai is an open-source framework for building ranking sites that Codex agents can fully research and maintain automatically. Friends often ask me which AI tools they should use for different tasks. I wanted a ranking I could trust and point them to, but many of the...

<https://segue.ai/>

VERI

Hybrid Analysis for Secure MCP Tool Use in LLM Agents

ArXiv CS.AI +1 source

80%

arXiv:2607.25297v1 Announce Type: cross Abstract: The rapid development of large language model (LLM) agents has enabled their broad adoption across diverse real-world tasks. To standardize interactions between LLM agents and external environments, Model Context Protocol (MCP)...

<https://arxiv.org/abs/2607.25297>

VERI

SAM3D-Guided Object-Centric Representation Alignment for Vision-Language-Action Models

ArXiv CS.AI +1 source

80%

arXiv:2607.25912v1 Announce Type: cross Abstract: Vision-Language-Action (VLA) models have shown strong potential for general robot manipulation, but most existing models rely on 2D visual-language backbones and lack fine-grained 3D understanding of target objects, especially...

<https://arxiv.org/abs/2607.25912>

VERI

Addressable Recall Compaction for Long Context-Window Control in AI Agents

ArXiv CS.AI +1 source

80%

arXiv:2607.25066v1 Announce Type: new Abstract: Long-horizon LLM agents accumulate reasoning traces, actions, and tool observations that can eventually exceed a model's fixed context window. Existing compaction methods address this limitation by discarding, summarizing, or...

<https://arxiv.org/abs/2607.25066>

CONFIRMED SIGNALS — 154 stories

CONF

Show HN: Formally verified 3D CSG: Trust 93 lines spec, not 1000 lines AI code

HackerNews 60% HN 108

To my knowledge, this is the first formally verified implementation of a 3D constructive solid geometry (CSG) operation: mesh intersection, implemented in Lean 4 and verified against a concise specification that pins down the surface of the resulting mesh exactly and guarantees...

<https://github.com/schildep/verified-3d-mesh-intersection>

CONF

Show HN: Ctrlb-decompose: Strip the noise from logs before sending to LLMs

HackerNews 50% HN 49

hi hn, i made a terminal for mac. you press a hotkey and it pops up over whatever window you're in, kinda like spotlight. press it again and it's gone. it saves and indexes all your claude code sessions, so you can search through them and pick up any one right where you left it...

<https://github.com/ctrlb-hq/ctrlb-decompose>

CONF

Show HN: Cynative - Read-only CLI in Go that explains your live infrastructure

HackerNews 50% HN 15

Hey, we built an open-source CLI in Go purposed to help you answer security questions across your cloud, code and runtime. It connects to your GitHub, GitLab, AWS, GCP, Azure & K8s using the credentials that are already in your shell, while limiting itself to read-only. The core...

<https://github.com/cynative/cynative>

CONF

Show HN: BrowserAct: Browser Layer for Your AI Agent

HackerNews 50% HN 14

No summary — see link for full article.

<https://github.com/browser-act/skills>

0 CONF

Show HN: Lean4 Datalog DSL Based on Google Zanzibar for AI Projects

HackerNews 50% HN 11

Google Zanzibar datalog lang lets you describe concepts and express how they are related. I generalize it to DSL you can use on Lean4 (and other languages) this lets you represent a knowledge base you can construct, store and evaluate, have it under git and improve without big...

<https://github.com/jagg-ix/zil-lean>

1 CONF

Show HN: Magpie, a CLI to Turn Your AI Agent into a Bookkeeper

HackerNews 50% HN 6

Hi HN, Announcing the release of Magpie, a CLI tool which turns your AI agent into an expert bookkeeper, based on Jaybase. (<https://news.ycombinator.com/item?id=48999936>) As someone who owns a small, tech business, I have very simple books. somewhere between 10-30 transactions...

<https://github.com/kyle-visner/magpie>

2 CONF

Show HN: Minions Army Harness - Small Version of Claude Tag

HackerNews 50% HN 3

No summary — see link for full article.

<https://github.com/baristaGeek/minions-army-harness>

3 CONF

Show HN: MetriQ - An AI fitness coach who supports you

HackerNews 50% HN 2

metriQ started as a spreadsheet for my own training. The numbers were fine, but a spreadsheet never tells you anything — it just holds what you typed. So I switched to asking ChatGPT. It could explain things, but it forgot my history, answered the same question differently on...

<https://metriq.fitness>

4 **CONF****Show HN: AiTextToHandwriting] - Text to Handwriting Converter****HackerNews** 50% HN 2

Turn typed text into handwritten pages with custom fonts, paper styles, layout controls, and PNG, PDF, or ZIP export—no account required.

<https://aitexttohandwriting.com/>

5 **CONF****Show HN: Vyne - Zapier for DeFi users, run from any LLM or the workflow builder****HackerNews** 50% HN 2

No summary — see link for full article.

<https://app.vyne.finance>

6 **CONF****Fish Audio raises \$52M seed to build AI voice models for creators and enterprises****TechCrunch AI** 50%

Since launching last year, the startup today has more than 8 million people using the open source or hosted version of its models, and now generates annual recurring revenue of \$21 million.

<https://techcrunch.com/2026/07/28/fish-audio-raises-50m-seed-to-build-ai-voice-models-f...>

7 **CONF****The Download: OpenAI's predictable hack, and an AI stock sell-off****MIT Tech Review** 50%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. OpenAI called the Hugging Face attack unprecedented. But we've been here before. —Will Douglas Heaven, senior AI editor Reading OpenAI's...

<https://www.technologyreview.com/2026/07/28/1140868/the-download-openai-hack-ai-stock-s...>

8 **CONF****AI leaders sign a statement asking the government to do something about automated AI****The Verge AI** 50%

Employees of OpenAI and Anthropic, as well as Google, Meta, Thinking Machines, Microsoft, Mistral, and other leading AI labs, have written a statement to the US government supporting a potential slowdown of sorts for frontier AI development - or at least a speed-up of global...

<https://www.theverge.com/ai-artificial-intelligence/972161/ai-leaders-us-government-ope...>

9 **CONF****AI's finally expensive enough to make Wall Street nervous****The Verge AI** 50%

It's earnings season, and investors got an unpleasant surprise from Google: an increase on its spending estimate, to as much as \$205 billion - from the last quarter's projection of up to \$190 billion. Even the lower end of Google's new projected range - \$195 billion - is much...

<https://www.theverge.com/ai-artificial-intelligence/972119/ai-stock-fall-google-capex>

0 **CONF****Smart rings are looking like my kind of AI gadget****The Verge AI** 50%

Over the last few months, I've spent a lot of time talking to my computer. One underrated feature of the LLM revolution has been a remarkable leap in all kinds of dictation technology - even the fastest, cheapest models are getting very good at understanding and processing...

<https://www.theverge.com/gadgets/971744/smart-ring-ai-gadget-stream-index>

1 CONF

Why China is giving away its best AI models

The Verge AI 50%

Silicon Valley has spent much of the past week on red alert, digesting the arrival of Moonshot AI's Kimi K3, a Chinese AI model that can allegedly beat some of the best systems built by US companies at a fraction of the cost. Its performance alone would have been enough to...

<https://www.theverge.com/ai-artificial-intelligence/971444/how-chinese-open-weight-ai-m...>

2 CONF

Do Models Fake Alignment Without Clear Consequences?

ArXiv CS.AI 50%

arXiv:2607.24758v1 Announce Type: new Abstract: Large language models are capable of recognizing evaluation contexts and altering their behavior to reflect evaluator expectations rather than typical deployment behaviors, a phenomenon known as alignment faking. The reasons why...

<https://arxiv.org/abs/2607.24758>

3 CONF

Beyond Memory: A Templated Substrate for Heterogeneous Collaborative Knowledge Work with LLM Agents

ArXiv CS.AI 50%

arXiv:2607.24759v1 Announce Type: new Abstract: Research projects, educational efforts, and adjacent knowledge work accumulate findings, decisions, and reasoning that future collaborators rarely recover. The parts most useful to that work, including dead ends and walked-back...

<https://arxiv.org/abs/2607.24759>

4 CONF

Kernel Forge: An Agent Harness for LLM-based Generation and Optimization of CUDA Kernels

ArXiv CS.AI 50%

arXiv:2607.24762v1 Announce Type: new Abstract: Machine learning models are increasingly embedded in everyday software, and most of their runtime is spent in a small set of compute kernels such as matrix multiplication, convolution, and normalization. Optimizing these kernels is...

<https://arxiv.org/abs/2607.24762>

5 CONF

Lantern: Conflict-Aware Gradient Blending for Physics-Guided Diffusion Models in Calorimeter Simulation

ArXiv CS.AI 50%

arXiv:2607.25060v1 Announce Type: cross Abstract: Monte Carlo simulation of calorimeter showers is a principal bottleneck for the High-Luminosity LHC, and diffusion models have emerged as fast, high-fidelity surrogates. Their denoising objective is purely statistical, however: a...

<https://arxiv.org/abs/2607.25060>

6 CONF

PATHFinder Agent for Tailored Prenatal Care

ArXiv CS.AI 50%

arXiv:2607.24768v1 Announce Type: new Abstract: Prenatal care is an important preventive service designed to improve outcomes for pregnant individuals. The American College of Obstetricians and Gynecologists (ACOG) recently introduced guidelines advocating tailored prenatal...

<https://arxiv.org/abs/2607.24768>

7 CONF

Measuring the State of Open Science in Transportation Using Large Language Models

ArXiv CS.AI 50%

arXiv:2601.14429v2 Announce Type: replace-cross Abstract: Open science initiatives have strengthened scientific integrity and accelerated research progress across many fields, but the state of their practice within transportation research remains under-investigated. Key features...

<https://arxiv.org/abs/2601.14429>

8 CONF

Atmospheric Diffusion-Guided Spatio-Temporal Transformer for Nuclear Radiation Forecasting

ArXiv CS.AI 50%

arXiv:2607.24774v1 Announce Type: new Abstract: Nuclear radiation, the energy released during atomic decay, poses persistent risks to public health and the environment, and concerns have only grown since the Fukushima accident and the recent commencement of treated-water...

<https://arxiv.org/abs/2607.24774>

9 CONF

Steering topology distributions for unified generative design of architected metamaterials

ArXiv CS.AI 50%

arXiv:2607.24777v1 Announce Type: new Abstract: Architected metamaterials derive their functions from structure, creating vast opportunities to program physical responses through topology design. However, existing design methods are often tailored to individual design problems,...

<https://arxiv.org/abs/2607.24777>

0 CONF

LivingArena: Do LLMs Know What Other LLMs Don't? Peer-Probing as Scalable Evaluation

ArXiv CS.AI 50%

arXiv:2607.24780v1 Announce Type: new Abstract: Evaluating frontier LLMs is challenging: static benchmarks suffer from contamination and saturation -- leaving users unable to distinguish top models and developers blind to specific failure modes -- while human preference is...

<https://arxiv.org/abs/2607.24780>

1 CONF

Localizing Persona Representations in LLMs

ArXiv CS.AI 50%

arXiv:2505.24539v4 Announce Type: replace-cross Abstract: We present a study on how and where personas -- defined by distinct sets of human characteristics, values, and beliefs -- are encoded in the representation space of large language models (LLMs). Using a range of dimension...

<https://arxiv.org/abs/2505.24539>

2 CONF

On the Use of LLMs for Specialised Terminology: A Good Alternative to Corpora?

ArXiv CS.AI 50%

arXiv:2607.24784v1 Announce Type: new Abstract: Specialised translation relies on the use of documentary and terminological resources, including corpora. These resources are particularly useful for terminology. However, their compilation and exploitation have several...

<https://arxiv.org/abs/2607.24784>

3 CONF

LLM as Forecasting Planner: Training-Free Text Conditioning for Time-Series Foundation Models

ArXiv CS.AI 50%

arXiv:2607.24892v1 Announce Type: cross Abstract: Text-conditioned time-series forecasting predicts a series from both its numerical history and natural-language context, allowing forecasts to account for events and constraints that the past alone cannot reveal. This requires...

<https://arxiv.org/abs/2607.24892>

4 CONF

GLIDE: Guided Layerwise Hybrid Attention for Efficient LLM Inference

ArXiv CS.AI 50%

arXiv:2607.24788v1 Announce Type: new Abstract: As Large Language Models scale to increasingly long contexts, the memory I/O and computational overhead of the Key-Value (KV) cache during decoding emerges as the primary throughput bottleneck. To address this, we propose GLIDE, a...

<https://arxiv.org/abs/2607.24788>

5 CONF

Reasoning with Memory: A Temporal Granularity-Adaptive Framework for Training-Free Long Video Understanding

ArXiv CS.AI 50%

arXiv:2607.24794v1 Announce Type: new Abstract: While Multimodal Large Language Models (MLLMs) demonstrate superior generalization in fundamental video tasks, restricted context windows limit their long video understanding. To accommodate this constraint, models typically resort...

<https://arxiv.org/abs/2607.24794>

6 CONF

RRS-10K: A Multitask Vision-Language Model Benchmark for Rare Remote Sensing Image Interpretation

ArXiv CS.AI 50%

arXiv:2607.24810v1 Announce Type: new Abstract: Vision-language models (VLMs) have achieved strong performance on general remote sensing tasks. However, their capability for rare scenes remains insufficiently understood, because existing benchmarks are dominated by common urban...

<https://arxiv.org/abs/2607.24810>

7 CONF

Explainable AI for Chronic Kidney Disease Prediction Using Simulated Federated Learning

ArXiv CS.AI 50%

arXiv:2607.25348v1 Announce Type: cross Abstract: Chronic Kidney Disease (CKD), characterized by the gradual loss of kidney function, remains a significant public health challenge. Early detection is crucial for preventing severe complications and enhancing patient outcomes. In...

<https://arxiv.org/abs/2607.25348>

8 CONF

TopoGR: Revealing and Preserving Latent Structure of Semantic ID in Generative Recommendation

ArXiv CS.AI 50%

arXiv:2607.25216v1 Announce Type: cross Abstract: Semantic ID-based generative recommendation tokenizes each item into a sequence of discrete semantic IDs and predicts the next item by generating semantic IDs. However, existing methods typically regard SIDs as independent...

<https://arxiv.org/abs/2607.25216>

9 CONF

Chart-Supported or Model-Supplied? Examining MLLM-Generated Claims for Accessible Visualization

ArXiv CS.AI 50%

arXiv:2607.25021v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) can connect visualization patterns to external causes, consequences, and domain knowledge, but the evidential basis of these interpretations is often unclear. We present an exploratory study...

<https://arxiv.org/abs/2607.25021>

0 CONF

Psychological Influences of Conversational AI: Research and Design Directions for Reducing Harm and Promoting Well-Being

ArXiv CS.AI 50%

arXiv:2607.25057v1 Announce Type: new Abstract: As conversational AI systems become increasingly integrated into daily life, their potential effects on user well-being require ongoing attention. While consumer-facing generalist models can provide benefits, including improved...

<https://arxiv.org/abs/2607.25057>

1 CONF

Similar Models Learn Differently: Final-Window Pretraining Shapes Post-Training Beyond SFT

ArXiv CS.AI 50%

arXiv:2607.25063v1 Announce Type: new Abstract: Developers judge a model checkpoint by how it behaves. After supervised fine-tuning (SFT), two checkpoints that perform about the same across relevant benchmarks are treated as interchangeable, equally ready for the next alignment...

<https://arxiv.org/abs/2607.25063>

2 CONF

How Often Should a Recommender Call an LLM? Value-Weighted Routing, Monitoring, and Seasonal Robustness

ArXiv CS.AI 50%

arXiv:2607.25068v1 Announce Type: new Abstract: Routing decisions between a cheap heuristic and an expensive large language model (LLM) are typically framed as a difficulty problem: send the hard cases to the expensive path. We argue this framing is incomplete because difficulty...

<https://arxiv.org/abs/2607.25068>

3 CONF

Matryoshka Agent: Unfolding Sub-Agents for Long-Horizon Machine Learning Engineering

ArXiv CS.AI 50%

arXiv:2607.25090v1 Announce Type: new Abstract: Machine learning engineering (MLE) tasks require long-horizon decision making over iterative solution debugging and refinement, under expensive and feedback-driven environment interactions. Developing and training a monolithic...

<https://arxiv.org/abs/2607.25090>

4 CONF

Image Quality Dependent Degradation for AI Systems

ArXiv CS.AI 50%

arXiv:2607.25736v1 Announce Type: cross Abstract: Perception is one of the primary applications where neural networks outperform conventional algorithms. One example is AI systems for automated driving, which can detect pedestrians based on image data and avoid them accordingly....

<https://arxiv.org/abs/2607.25736>

5 CONF

Less Data, Better Alignment: Data-Centric Multi-Evaluator Agreement for Preference Optimization

ArXiv CS.AI 50%

arXiv:2607.25136v1 Announce Type: new Abstract: Research on preference optimization often varies the training objective while holding the data fixed. We instead ask whether a small, high-confidence set of on-policy responses can provide a reliable learning signal. Our method,...

<https://arxiv.org/abs/2607.25136>

6 CONF

Cyber-Capable AI Agents: Vulnerabilities, Evaluation Containment, and Defensive Response

ArXiv CS.AI 50%

arXiv:2607.25379v1 Announce Type: new Abstract: Cyber-capable AI agents combine language models with tools, memory, and execution environments to perform multi-step offensive-security tasks. Existing work separately measures cyber capability and catalogs attacks against agent...

<https://arxiv.org/abs/2607.25379>

7 CONF

When Do Agent Loops Mistake Stagnation for Progress? Self-Evaluation Bias and Externally Grounded Verification in Long-Running Autonomous LLM Agent Loops

ArXiv CS.AI 50%

arXiv:2607.25152v1 Announce Type: new Abstract: Long-running autonomous agents plan, act, and judge their own completion without human intervention. When an agent grades its own work, self-evaluation bias takes hold: plausible changes are accepted as progress while real-world...

<https://arxiv.org/abs/2607.25152>

8 CONF

Observing sycophantic AI validate others reduces its appeal but not its persuasiveness

ArXiv CS.AI 50%

arXiv:2607.25166v1 Announce Type: new Abstract: AI chatbots can be "sycophantic," or overly agreeable and flattering toward users. Sycophantic AI has been shown to entrench attitudes, yet users frequently fail to recognize it (a phenomenon we call "sycophancy blindness"). We...

<https://arxiv.org/abs/2607.25166>

9 CONF

Beyond Counts: A Distributional Robustness Margin For Pathology Foundation Models

ArXiv CS.AI 50%

arXiv:2607.25497v1 Announce Type: cross Abstract: Pathology foundation models are approaching clinical deployment, yet remain vulnerable to systematic non-biological variation across centres. Differences in tissue preparation, staining and scanning are strongly encoded in their...

<https://arxiv.org/abs/2607.25497>

10 CONF

Many-body Tipping Dynamics of ChatGPT-like AIs

ArXiv CS.AI 50%

arXiv:2607.25279v1 Announce Type: new Abstract: Why do ChatGPT-like AIs, despite major architectural and training differences, unexpectedly tip to undesirable content (e.g. harmful, misleading, repetitive) even under deterministic greedy decoding? We show that a broad class of...

<https://arxiv.org/abs/2607.25279>

1 **CONF****Instruction-Tuned Language Models Cannot Sample from Distributions They Can Describe****ArXiv CS.AI** **50%**

arXiv:2607.25292v1 Announce Type: new Abstract: Silicon sampling uses language models as proxies for human survey respondents, treating each model call as an independent draw from the persona's response distribution. We show this draw does not exist: instruction-tuned models do...

<https://arxiv.org/abs/2607.25292>

2 **CONF****From Cellular Responses to Pharmacological Domains: Multimodal Zero-Shot Drug Representation Learning****ArXiv CS.AI** **50%**

arXiv:2607.25322v1 Announce Type: new Abstract: Multimodal drug discovery enables drug representation learning beyond chemical structure by incorporating cellular responses such as gene expression and cell morphology. However, direct fusion and instance-level contrastive...

<https://arxiv.org/abs/2607.25322>

3 **CONF****Dual-Domain Manifold Modeling for Hyperspectral Image Fusion****ArXiv CS.AI** **50%**

arXiv:2607.25338v1 Announce Type: new Abstract: Achieving a coherent integration of spectral richness and spatial fidelity remains a central objective in hyperspectral image fusion. However, existing hyperspectral image fusion methods struggle to effectively model geometric...

<https://arxiv.org/abs/2607.25338>

4 **CONF****Explanation-Bound Tool Execution for AI Agents: Server-Verified Action Claims Without Trusting Model Rationales****ArXiv CS.AI** **50%**

arXiv:2607.25364v1 Announce Type: new Abstract: Tool-using agents expose structured calls but commonly attach free-form rationales. Such rationales are neither authorization nor reliable introspection. We present Explanation-Bound Tool Execution (EBTE), a claim-carrying...

<https://arxiv.org/abs/2607.25364>

5 **CONF****AI Deployment and Cyber Governance Failures in Public-Sector Organizations: A Typological Analysis****ArXiv CS.AI** **50%**

arXiv:2607.25368v1 Announce Type: new Abstract: The intersection of artificial intelligence adoption, cybersecurity governance, and public sector institutional constraints has not been examined as a unified analytical problem in the existing literature. Studies address AI...

<https://arxiv.org/abs/2607.25368>

6 **CONF****HANDBOOK.md: A Benchmark for Long-Context Agentic Instruction Following****ArXiv CS.AI** **50%**

arXiv:2607.25398v1 Announce Type: new Abstract: Language-model agents are increasingly deployed under standing instructions: a system prompt, a policy file, or a skills document is placed in context, and the agent is trusted to let it govern every action that follows. Existing...

<https://arxiv.org/abs/2607.25398>

- 7 **CONF**
Context Assembly as the Controlled Variable: A Control-Theoretic View of Harness Policies for Frozen LLM Agents
 ArXiv CS.AI 50%
 arXiv:2607.25408v1 Announce Type: new Abstract: A growing body of 2026 work applies control theory to LLM agents: Lyapunov-certified stability for tool-mediated controllers (Prinos et al., "Stable Agentic Control", 2026), sample-complexity bounds for sparse policies over massive...
<https://arxiv.org/abs/2607.25408>
- 8 **CONF**
A Control System, a Dataset, and a Recipe for Making Frozen LLM Agents Learn a Domain
 ArXiv CS.AI 50%
 arXiv:2607.25415v1 Announce Type: new Abstract: Production LLM agents are increasingly assembled from a frozen model wrapped in a harness: a prompt template, a tool set, a memory/retrieval layer, a planning strategy, and a verification policy. Two 2026 systems, Meta-Harness (Lee...
<https://arxiv.org/abs/2607.25415>
- 9 **CONF**
Salient Knowledge Pathways: Sparse Cross-Modal Routing for Efficient Knowledge-Intensive Multimodal Question Answering
 ArXiv CS.AI 50%
 arXiv:2607.25422v1 Announce Type: new Abstract: Knowledge-intensive multimodal question answering (KI-MMQA) sits at the intersection of three expensive primitives: long visual token sequences, dense retrieval over large external corpora, and full cross-modal fusion. Existing...
<https://arxiv.org/abs/2607.25422>
- 0 **CONF**
Early Detection of Distributed Backdoors in Multi-Agent LLM Systems: A Characterization Study
 ArXiv CS.AI 50%
 arXiv:2607.24893v1 Announce Type: cross Abstract: Multi-agent LLM systems can be attacked by a payload that no single agent ever holds in full: a poisoned tool hides encrypted fragments in its observations, spreads them across several agents, and an external step reassembles and...
<https://arxiv.org/abs/2607.24893>
- 1 **CONF**
Balancing multiscale similarity and cartographic constraints: A similarity-driven optimization framework for line generalization
 ArXiv CS.AI 50%
 arXiv:2607.25474v1 Announce Type: new Abstract: Cartographic generalization is essential for generating multiscale map representations by balancing information preservation and cartographic readability. However, automated generalization remains challenging because existing...
<https://arxiv.org/abs/2607.25474>
- 2 **CONF**
PatientAgentBench: A Benchmark Framework for Evaluating Patient-Facing Health AI Agents
 ArXiv CS.AI 50%
 arXiv:2607.25485v1 Announce Type: new Abstract: Health AI is evolving from answering questions to agentic systems that converse with patients, reason about health records, and act on their behalf. Primary care guards against diagnostic errors and unsafe care; agents assisting in...
<https://arxiv.org/abs/2607.25485>

3 CONF

CoTinyVLA: Chain-of-Thought Distillation for a Sub-Billion-Parameter Vision-Language-Action Model

ArXiv CS.AI 50%

arXiv:2607.25487v1 Announce Type: new Abstract: Vision-Language-Action (VLA) models translate natural-language commands into robot action sequences, but leading systems on the LIBERO-Plus robustness benchmark use three- to seven-billion-parameter backbones whose memory demands...

<https://arxiv.org/abs/2607.25487>

4 CONF

Are the High-weight Neurons the Important Ones in Image Classification Neural Networks?

ArXiv CS.AI 50%

arXiv:2607.25529v1 Announce Type: new Abstract: As neural network models for image classification advance, neurons play critical roles in pruning, backdoor defense, and interpretability. Yet existing work lacks clarity on the weight-importance relationship. We address this with...

<https://arxiv.org/abs/2607.25529>

5 CONF

From Training to Deployment: Post-Hoc Causal Feature Identification via Sensitivity Ratios

ArXiv CS.AI 50%

arXiv:2607.25546v1 Announce Type: new Abstract: Given a model that is already trained, which features does it rely on causally versus spuriously? Existing methods require access to the training procedure and cannot answer this post-hoc. We introduce the \textbf{Normalised...}

<https://arxiv.org/abs/2607.25546>

6 CONF

Distilling Temporal Search and Reasoning: Evolving LLMs for Future Prediction via Harness-Assisted Efficient Data Synthesis

ArXiv CS.AI 50%

arXiv:2607.25554v1 Announce Type: new Abstract: Future event prediction carries broad social impact yet remains challenging. SOTA approaches augment LLMs with external agent frameworks whose predictive capability vanishes once the harness is removed. While recent Tool-Integrated...

<https://arxiv.org/abs/2607.25554>

7 CONF

Multi-Sensor Alignment for Weather Simulations

ArXiv CS.AI 50%

arXiv:2607.25612v1 Announce Type: new Abstract: Perception tasks for autonomous vehicles need to work satisfactorily in adverse weather conditions. Due to lack of real-world weather datasets, weather simulations are a promising alternative. To ensure simulations closely mirror...

<https://arxiv.org/abs/2607.25612>

8 CONF

Beyond Epistemia: Epistemic Schizologia and Large Language Models as Techno-Semiotic Machines

ArXiv CS.AI 50%

arXiv:2607.25620v1 Announce Type: new Abstract: Quattrociocchi and colleagues warn that the fluent outputs of large language models may allow linguistic plausibility to substitute for epistemic evaluation, producing the condition they call *Epistemia*: the experience of...

<https://arxiv.org/abs/2607.25620>

9 CONF

Joint Text-Audio Alignment for EEG-to-Text Decoding in Chinese Speech Production and Perception

ArXiv CS.AI 50%

arXiv:2607.25626v1 Announce Type: new Abstract: Decoding speech information directly from scalp electroencephalography (EEG) into text provides a potential non-invasive neural communication pathway for individuals with severe speech and motor impairments. Compared with invasive...

<https://arxiv.org/abs/2607.25626>

0 CONF

AlriskEval-edu Demo: Auditing of Pedagogical Risks in Educational Explanations

ArXiv CS.AI 50%

arXiv:2607.25634v1 Announce Type: new Abstract: We present AlriskEval-edu Demo, a platform that audits the pedagogical quality of instructional explanations and provides explainable audit results. The platform evaluates an explanation against a rubric covering five dimensions of...

<https://arxiv.org/abs/2607.25634>

1 CONF

Localized Adaptation Reveals Distinct Learning Signatures in Transformers

ArXiv CS.AI 50%

arXiv:2607.25663v1 Announce Type: new Abstract: Transformer adaptation is typically distributed across model depth, even when the intended change is narrow. We investigate how adaptation site shapes what a model learns, how well that learning generalizes, and how selectively it...

<https://arxiv.org/abs/2607.25663>

2 CONF

OmniDelta: Skill-Driven Budget Allocation for Token Compression in OmniLLMs

ArXiv CS.AI 50%

arXiv:2607.25669v1 Announce Type: new Abstract: Emerging Omni-modal Large Language Models (OmniLLMs) enable unified understanding of text, audio, and video, but their long audio-video token sequences introduce substantial memory and inference costs. Existing compression methods...

<https://arxiv.org/abs/2607.25669>

3 CONF

Cognivia: A Cognitive Behavioral Therapy Copilot for Evidence-Based Mental Healthcare

ArXiv CS.AI 50%

arXiv:2607.25681v1 Announce Type: new Abstract: Cognitive distortion amplifies negative emotions and contributes to mental health disorders. Cognitive Behavioral Therapy (CBT) is an effective way to address cognitive distortions, but its large-scale application is limited by the...

<https://arxiv.org/abs/2607.25681>

4 CONF

Nudging Sustainable Choices through LLM-Generated Recommendation Explanations

ArXiv CS.AI 50%

arXiv:2607.25726v1 Announce Type: new Abstract: Recommender systems mediate everyday consumption, offering a promising channel for encouraging sustainable choices. Prior research shows that explanations influence users' perceptions of recommendations and can support more...

<https://arxiv.org/abs/2607.25726>

5 CONF

Distributed Constraint Optimization via Online Learning and Iterative Pricing with Application to Large-Scale Satellite Scheduling

ArXiv CS.AI 50%

arXiv:2607.25835v1 Announce Type: new Abstract: Distributed constraint optimization problems (DCOPs) provide a popular framework for distributed decision making under limited communication, but many real-world instances are too large to solve monolithically. We address this...

<https://arxiv.org/abs/2607.25835>

6 CONF

HiSkill: Empowering LLM Agents with Hierarchical Skill Graphs

ArXiv CS.AI 50%

arXiv:2607.25853v1 Announce Type: new Abstract: Skills have become an important abstraction for enabling large language model (LLM) agents to reuse past experience in long-horizon interactive tasks. However, existing trajectory-to-skill methods often produce flat collections of...

<https://arxiv.org/abs/2607.25853>

7 CONF

Messier: A High-Resolution Corpus for Cross-Benchmark Agent Evaluation

ArXiv CS.AI 50%

arXiv:2607.25891v1 Announce Type: new Abstract: Evaluating AI agents in interactive environments is hindered by fragmented tasks, scaffolds, verifiers, and scoring rules. Existing efforts focus on narrow settings, remain limited in scale, or require costly reruns, leaving much...

<https://arxiv.org/abs/2607.25891>

8 CONF

dtControl2+\$\varepsilon\$: Trading Optimality for Explainability in MDPs via Decision Trees

ArXiv CS.AI 50%

arXiv:2607.25925v1 Announce Type: new Abstract: Over the past decade, decision trees have been used to represent controllers (a.k.a. policies) in an explainable way, with dtControl2 as a current state-of-the-art tool. However, for systems that are large or have many corner...

<https://arxiv.org/abs/2607.25925>

9 CONF

A Cost-Effective Multimodal LLM Reasoning Framework for Question Answering over Irregular Clinical Time Series

ArXiv CS.AI 50%

arXiv:2607.25947v1 Announce Type: new Abstract: Question answering (QA) over irregular clinical time series (ICTS) plays a pivotal role in a wide range of healthcare applications. Although recent multimodal time-series large language models (LLMs) have shown considerable promise...

<https://arxiv.org/abs/2607.25947>

0 CONF

CHARM: A Multimodal Graph Foundation Model with Hierarchical Context Modeling for Zero-Shot Transfer

ArXiv CS.AI 50%

arXiv:2607.26023v1 Announce Type: new Abstract: Graph foundation models (GFMs) have emerged as a promising paradigm for transferring knowledge across graph domains and tasks. Real-world graphs associate nodes with text, images, and other modalities, making multimodal graphs...

<https://arxiv.org/abs/2607.26023>

1 **CONF****Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment**

ArXiv CS.AI 50%

arXiv:2607.26034v1 Announce Type: new Abstract: Technological races create tension between speed and safety: actors may gain by moving faster than competitors, even when risky development is harmful. This is prominent in debates about artificial intelligence (AI), where...

<https://arxiv.org/abs/2607.26034>

2 **CONF****Domain-Prior-Regularized Graph Modeling for Anomaly Detection in Cyber-Physical Systems**

ArXiv CS.AI 50%

arXiv:2607.23197v1 Announce Type: cross Abstract: Anomaly detection on multivariate sensor time series is critical for industrial monitoring of cyber-physical systems (CPS), where even subtle deviations from normal behavior can indicate process disruption. Recent graph-based...

<https://arxiv.org/abs/2607.23197>

3 **CONF****Neural Network Learning of One-Bit Protocols for Qubit Measurement Simulation**

ArXiv CS.AI 50%

arXiv:2607.23645v1 Announce Type: cross Abstract: Communication complexity provides a natural framework for quantifying the classical resources required to reproduce quantum statistics. In the qubit prepare-and-measure scenario, two classical bits have been shown to be necessary...

<https://arxiv.org/abs/2607.23645>

4 **CONF****DocAnnot -- Accelerating the Creation of Key Information Extraction Datasets with GenAI-Powered Auto-annotation**

ArXiv CS.AI 50%

arXiv:2607.24745v1 Announce Type: cross Abstract: Key Information Extraction (KIE) is vital for many document applications, but creating training datasets is traditionally a time-consuming manual process. We introduce DocAnnot, a framework that significantly accelerates KIE...

<https://arxiv.org/abs/2607.24745>

5 **CONF****VLD-RAG: Agentic Vision-Language Retrieval-Augmented Generation for Long, Visually-Rich Multi-Page Documents**

ArXiv CS.AI 50%

arXiv:2607.24748v1 Announce Type: cross Abstract: Visually-rich documents such as reports, slides, and manuals often distribute the evidence needed to answer a question across multiple pages, mixing text with layout cues, tables, charts, and figures. This work studies multimodal...

<https://arxiv.org/abs/2607.24748>

6 **CONF****Game AI Not Fun? A Scoping Review and Meta-Analysis on the Differences in Enjoyment between Human and Computer Opponents**

ArXiv CS.AI 50%

arXiv:2607.24749v1 Announce Type: cross Abstract: Although advancements in game character AI aim to enhance player engagement, evidence suggests that perceiving an opponent as artificial can diminish the psychological experience. This paper presents a scoping review and...

<https://arxiv.org/abs/2607.24749>

7 CONF

CARE-MH: Towards Unified, Reproducible, and Comparable Evaluation of Mental Health LLMs

ArXiv CS.AI 50%

arXiv:2607.24754v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used to provide mental health support, requiring reliable evaluation of safety, empathy, and therapeutic appropriateness. However, existing mental health benchmarks are difficult to...

<https://arxiv.org/abs/2607.24754>

8 CONF

Patterns of Learner-AI Interaction and Academic Performance in an Object-Oriented Programming Course

ArXiv CS.AI 50%

arXiv:2607.24755v1 Announce Type: cross Abstract: This full research paper examines how different forms of learner-AI interaction relate to learning outcomes in object-oriented programming (OOP) courses. Generative artificial intelligence (GenAI) tools are increasingly used by...

<https://arxiv.org/abs/2607.24755>

9 CONF

Measuring and Improving Behavioral Consistency in Large Language Models through Fact-Heuristic-Emotion State Enforcement

ArXiv CS.AI 50%

arXiv:2607.24765v1 Announce Type: cross Abstract: Large language models (LLMs) can give different answers to the same decision problem across runs, and reverse a decision when their own prior answer returns as context. We ask whether this instability can be measured and...

<https://arxiv.org/abs/2607.24765>

0 CONF

Three Sides of Retrieval: Factorial Evidence for Document-Side, Query-Side, and Answer-Side Complementarity in RAG

ArXiv CS.AI 50%

arXiv:2607.24781v1 Announce Type: cross Abstract: RAG systems rely on chunking, which destroys structural information in documents. Existing heading-based retrieval (Jeong et al., 2025) requires multiple LLM calls per document and returns sub-chunks within matched sections. We...

<https://arxiv.org/abs/2607.24781>

1 CONF

From Naive RAG to Deep Agentic Retrieval: An Evolving Context Engineering Pipeline for Regulatory Compliance

ArXiv CS.AI 50%

arXiv:2607.24791v1 Announce Type: cross Abstract: Retrieval-augmented generation (RAG) is the dominant paradigm for applying large language models (LLMs) to enterprise document corpora, yet naive implementations encounter hard limits as corpus scale and query complexity grow....

<https://arxiv.org/abs/2607.24791>

2 CONF

AI-Assisted Knowledge Access for Legacy Enterprise Asset Management in Energy Operations: A Practical Retrieval System

ArXiv CS.AI 50%

arXiv:2607.24792v1 Announce Type: cross Abstract: Energy utilities still run engineering work management, engineering procurement, and inventory processes on long-lived enterprise asset management platforms. Replacing these platforms is often cost prohibitive and operationally...

<https://arxiv.org/abs/2607.24792>

3 CONF

Selective Impairment of Motor Recovery from Typing Errors in Parkinson's Disease: A Survival Analysis

ArXiv CS.AI 50%

arXiv:2607.24796v1 Announce Type: cross Abstract: Parkinson's disease (PD) affects multiple, dissociable stages of motor and cognitive control. We ask whether passively-collected keystroke dynamics can distinguish two of these stages: noticing a self-generated error (error...

<https://arxiv.org/abs/2607.24796>

4 CONF

Artificial Epanorthosis: Why large language models overuse a classical rhetorical figure, and how to mitigate it

ArXiv CS.AI 50%

arXiv:2607.21498v2 Announce Type: replace-cross Abstract: A rhetorical figure that Cicero and Quintilian catalogued two thousand years ago reappears, systematically, in the text of large language models: epanorthosis, the self-correction of the specimen {\guillemotleft}This is...

<https://arxiv.org/abs/2607.21498>

5 CONF

Retrieval-Augmented Generation in LLMs for Mental Health: Quantifying the Incremental Contribution of Retrieval Within a Layered Safety Architecture

ArXiv CS.AI 50%

arXiv:2607.24817v1 Announce Type: cross Abstract: Digital mental health interventions (DMHIs) offer scalable support, but ensuring they accurately detect users' intent during volatile situations can be challenging. Pure parametric Large Language models (LLMs) do not contain...

<https://arxiv.org/abs/2607.24817>

6 CONF

Dual-Level Atomic and Coordination Geometry Learning for Crystal Property Prediction Using Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2607.24818v1 Announce Type: cross Abstract: Accurate prediction of crystal properties remains a key challenge in computational materials science. While graph neural networks (GNNs) such as CGCNN, MEGNet, ALIGNN, and SchNet have shown strong performance, they primarily...

<https://arxiv.org/abs/2607.24818>

7 CONF

Foundation Models for EEG Are Blind to Long-Range Temporal Correlations: A Spectral-Temporal Dissociation Behind Their Cross-Population Fragility

ArXiv CS.AI 50%

arXiv:2607.24834v1 Announce Type: cross Abstract: Objective. Electroencephalography (EEG) foundation models (FMs) are trained to reconstruct or contrastively align short patches, then pooled into a fixed embedding. We tested whether these embeddings retained the long-range...

<https://arxiv.org/abs/2607.24834>

8 CONF

MedJudgeRAG: Option-Wise Evidence Judgment with Dynamic Knowledge Graphs for Medical MCQA

ArXiv CS.AI 50%

arXiv:2607.24838v1 Announce Type: cross Abstract: In medical multiple-choice question answering (MCQA), Retrieval-Augmented Generation (RAG) can supplement the domain knowledge of language models (LMs). However, since vanilla RAG indiscriminately utilizes retrieved documents, it...

<https://arxiv.org/abs/2607.24838>

9 CONF

Extremal Chowla sets and their linear analogues: A human-AI mathematical investigation using Co-Scientist

ArXiv CS.AI 50%

arXiv:2607.24847v1 Announce Type: cross Abstract: We introduce an extremal invariant associated with Chowla-type order conditions in finite groups. A nonempty subset S of a finite group G is called a Chowla set if every element of S has order greater than $|S|$, and we...

<https://arxiv.org/abs/2607.24847>

00 CONF

DisasterTD: Disaster Toponym Disambiguation Using Multimodal LLMs and Cross-View Geolocalization

ArXiv CS.AI 50%

arXiv:2607.24856v1 Announce Type: cross Abstract: Social media imagery (SMI) provides timely and fine-grained ground perspectives that are valuable for situational awareness and emergency response. Unlike satellite or aerial imagery, SMI can capture disaster impacts and...

<https://arxiv.org/abs/2607.24856>

01 CONF

Tokens are All You Need: Dual-purpose Semantic IDs for Achieving LLM-Level I/O Efficiency in recommendation systems

ArXiv CS.AI 50%

arXiv:2607.24865v1 Announce Type: cross Abstract: Large-scale recommendation systems face "Memory Wall" bottlenecks due to massive, dense embedding tables. While generative retrieval uses discrete tokens for IDs, high-dimensional context still relies on inefficient dense...

<https://arxiv.org/abs/2607.24865>

02 CONF

GraphRareBench: An Auditable Graph-Evidence Benchmark for Phenotype-Driven Rare-Disease Diagnosis

ArXiv CS.AI 50%

arXiv:2607.24878v1 Announce Type: cross Abstract: Phenotype-driven diagnostic benchmarks usually report the rank of the reference disease, but they rarely reveal which plausible alternatives are ranked above it or what evidence a tool-using model examines before making its...

<https://arxiv.org/abs/2607.24878>

03 CONF

From Dyad to Triad: Eliciting XAI Requirements in Stroke Rehabilitation

ArXiv CS.AI 50%

arXiv:2607.25423v1 Announce Type: cross Abstract: Eliciting explainable AI (XAI) requirements from stroke survivors presents a methodological challenge with direct implications for the design of trustworthy brain-computer interfaces for rehabilitation. How can patients and...

<https://arxiv.org/abs/2607.25423>

04 CONF

Generative Distributionally Robust Optimization

ArXiv CS.AI 50%

arXiv:2607.24983v1 Announce Type: cross Abstract: Generative models are increasingly adopted in distributionally robust optimization (DRO), but existing approaches trade off model compatibility and adversarial structure: methods that accept arbitrary samplers do not restrict...

<https://arxiv.org/abs/2607.24983>

05 CONF

Preliminary Guidelines for Using and Evaluating GenAI Tools to Support Systematic Literature Reviews

ArXiv CS.AI 50%

arXiv:2607.24991v1 Announce Type: cross Abstract: Context: Generative AI (GenAI) and Large Language Models (LLMs) are increasingly used for academic tasks in software engineering and beyond, including systematic literature reviews (SLRs). However, while capable of summarizing...

<https://arxiv.org/abs/2607.24991>

06 CONF

Calibrated Partial Resets: Preventing Policy Collapse in Continual Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2607.24996v1 Announce Type: cross Abstract: Neural networks are hindered by accumulating dormant neurons and loss of expressivity throughout training, particularly in non-stationary data settings, such as continual supervised and reinforcement learning. Recently, neuron...

<https://arxiv.org/abs/2607.24996>

07 CONF

DS@GT ARC at CheckThat! 2026: LLM-Based Trace Ranking and Grouped Reward Modeling for Multilingual Numerical Claim Verification

ArXiv CS.AI 50%

arXiv:2607.25069v1 Announce Type: cross Abstract: Automated verification of numerical claims is a challenging problem, as it requires both language understanding and quantitative reasoning. This paper describes our system for CLEF 2026 CheckThat! Task 2, which focuses on ranking...

<https://arxiv.org/abs/2607.25069>

08 CONF

Learning from 53.6K Real-World Developer Edits of AI-Generated Code

ArXiv CS.AI 50%

arXiv:2607.25130v1 Announce Type: cross Abstract: Imperfections in AI-generated code require that software developers modify the generated code manually, or by re-prompting an AI programming assistant. Manual code edits provide more realistic and granular information on editing...

<https://arxiv.org/abs/2607.25130>

09 CONF

Agentic AI for Scientific Reasoning in Autonomous Quantum Sensing Experiments

ArXiv CS.AI 50%

arXiv:2607.25145v1 Announce Type: cross Abstract: We implement an agentic AI workflow built around a large language model (LLM) agent for autonomous experiments with nitrogen-vacancy (NV) centers in diamond. NV centers are a widely used platform for quantum sensing, and the...

<https://arxiv.org/abs/2607.25145>

10 CONF

RIDGE: An Autonomous Framework for Validation and Method Discovery in LLM-Generated Option Pricing

ArXiv CS.AI 50%

arXiv:2607.25199v1 Announce Type: cross Abstract: Automated code generation is becoming an important tool in quantitative finance, where large language models can generate option pricing implementations directly from mathematical model specifications. Validating such...

<https://arxiv.org/abs/2607.25199>

11 CONF

Laplace-PSN-IRT: Uncertainty Quantification for Neural Item Response Theory Models of LLM Benchmarks

ArXiv CS.AI 50%

arXiv:2607.25257v1 Announce Type: cross Abstract: Item Response Theory (IRT) has recently been proposed as a framework for evaluating large language model (LLM) benchmarks by separating a model's latent ability from the properties of individual benchmark items. Existing neural...

<https://arxiv.org/abs/2607.25257>

12 CONF

Bridging Compute- and Data-Optimal Pretraining

ArXiv CS.AI 50%

arXiv:2607.25271v1 Announce Type: cross Abstract: Classical compute-optimal scaling laws assume an unbounded supply of fresh pretraining data, yet pretraining is increasingly entering a regime in which compute grows faster than the availability of high-quality data. We propose...

<https://arxiv.org/abs/2607.25271>

13 CONF

ScaleResfusion: Residual Rectified Flow based on Residual Vector Field

ArXiv CS.AI 50%

arXiv:2607.25275v1 Announce Type: cross Abstract: Real-world Image Restoration (Real-IR) aims to recover high-quality (HQ) images from complex and unknown degradations. Although recent diffusion-based methods have substantially improved perceptual quality, their current designs...

<https://arxiv.org/abs/2607.25275>

14 CONF

CLBench-V: Evaluating Multimodal Context Learning from Grounding to Knowledge Acquisition

ArXiv CS.AI 50%

arXiv:2607.25294v1 Announce Type: cross Abstract: Real-world tasks often require models to learn from task-specific context rather than relying only on pre-trained knowledge. While recent work has highlighted this capability as context learning, existing evaluations mainly focus...

<https://arxiv.org/abs/2607.25294>

15 CONF

Retraction-Free Optimization over the Stiefel Manifold for the LoRA Fine-Tuning

ArXiv CS.AI 50%

arXiv:2607.25299v1 Announce Type: cross Abstract: Optimization over the Stiefel manifold plays a significant role in various machine learning tasks. Existing methods either use the retraction operators, requiring costly orthonormalization for large-scale matrices, or employ...

<https://arxiv.org/abs/2607.25299>

16 CONF

CAST: Game Solvers as Turn-Level Teachers for LLM Agents

ArXiv CS.AI 50%

arXiv:2607.25308v1 Announce Type: cross Abstract: Training large language models (LLMs) to act in long-horizon games is a promising step toward generalist decision-making, yet reinforcement learning with verifiable rewards (RLVR) relies on sparse final rewards that reveal little...

<https://arxiv.org/abs/2607.25308>

17 CONF

Every Time I Hire a Linguist, Inference Costs Go Down: On Linguistic Rules as Effective Prompt Compressors

ArXiv CS.AI 50%

arXiv:2607.25335v1 Announce Type: cross Abstract: Prompt compression shortens LLM input to reduce inference cost, yet existing methods score token importance through LM forward passes. It remains questionable whether such nuanced, costly token selection is necessary. Compression...

<https://arxiv.org/abs/2607.25335>

18 CONF

Data Quality Profiling at Scale with Progressive Sampling: A Benchmark for Data-Centric AI Pipelines

ArXiv CS.AI 50%

arXiv:2607.25356v1 Announce Type: cross Abstract: Data quality profiling -- computing missing-value rates, duplicate fractions, outlier densities, and functional-dependency violations -- is foundational for data-centric AI pipelines, yet exhaustive scans over millions of rows...

<https://arxiv.org/abs/2607.25356>

19 CONF

Rethinking Likelihood distributions: Student's t Likelihood Boosts Bayesian Neural Network Performance

ArXiv CS.AI 50%

arXiv:2607.25376v1 Announce Type: cross Abstract: In Bayesian neural networks (BNNs), variational inference is a widely adopted framework for modeling uncertainty in a distributional way, with the evidence lower bound (ELBO) serving as the standard objective function. Several...

<https://arxiv.org/abs/2607.25376>

20 CONF

Seen, Said, or Forgotten? A Causal Audit of Visual KV Memory Across Dialog Turns

ArXiv CS.AI 50%

arXiv:2607.25467v1 Announce Type: cross Abstract: Stateful multimodal assistants encode an image once but may answer questions about it many turns later. Attention-guided visual-KV eviction assumes that evidence irrelevant now will remain dispensable, although future questions...

<https://arxiv.org/abs/2607.25467>

21 CONF

At-the-Roofline Sparse Tensor Contractions on Vector Processors for Transformer Inference

ArXiv CS.AI 50%

arXiv:2607.25504v1 Announce Type: cross Abstract: Fine-grained weight pruning and activation sparsification have emerged as effective approaches for reducing the compute and memory cost of inference for Transformer models. In the moderate-sparsity regime, Gustavson's dataflow...

<https://arxiv.org/abs/2607.25504>

22 CONF

I2VShield: An Efficient Proactive Defense Framework against DiT-based Image-to-Video Models

ArXiv CS.AI 50%

arXiv:2607.25522v1 Announce Type: cross Abstract: The rapid advancement of video generation models has led to the increasing misuse of image-to-video (I2V) models. Although substantial progress has been made in detecting AI-generated videos, proactive defenses against I2V models...

<https://arxiv.org/abs/2607.25522>

23 CONF

IRIS: Reusable Identity Representations from Frozen LLMs for Entity Alignment

ArXiv CS.AI 50%

arXiv:2607.25579v1 Announce Type: cross Abstract: Entity alignment (EA) identifies entities across knowledge graphs (KGs) that refer to the same real-world object. Conventional EA methods mainly exploit explicit graph structures and textual fields, which often provide...

<https://arxiv.org/abs/2607.25579>

24 CONF

A Human-in-the-Loop Corpus for LLM-Based Simplification of Scientific Summaries

ArXiv CS.AI 50%

arXiv:2607.25630v1 Announce Type: cross Abstract: Interdisciplinary research is accelerating, yet scientific papers remain difficult to understand outside their home fields. We study large language model (LLM)-based simplification of scientific texts and present a...

<https://arxiv.org/abs/2607.25630>

25 CONF

Construction-Driven Injection: Linguistically-Grounded Edit-Based Code-Mixing Fingerprints for Large Language Models

ArXiv CS.AI 50%

arXiv:2607.25633v1 Announce Type: cross Abstract: Large language models (LLMs) are costly intellectual assets that remain exposed to unauthorized redistribution and commercial misuse. Injected fingerprints, i.e., trigger-target pairs embedded in model behavior, offer a...

<https://arxiv.org/abs/2607.25633>

26 CONF

F(AI)2R: Who Did What, and Who Checked? Verifiable AI Provenance as an Executable Skill

ArXiv CS.AI 50%

arXiv:2607.25637v1 Announce Type: cross Abstract: F(AI)2R is FAIR research with AI in the loop, twice: an AI-assisted authoring pass and a machine-readable audit pass over every artefact. AI systems now draft, refactor, and verify research artefacts, yet their contributions are...

<https://arxiv.org/abs/2607.25637>

27 CONF

Moral Hazard in Multi-Agent Language Models

ArXiv CS.AI 50%

arXiv:2607.23982v2 Announce Type: replace-cross Abstract: Cooperation can fail when socially valuable effort is costly, weakly observable, and mainly benefits others. Drawing on Holmström's team moral-hazard model, we introduce the Dialogue Moral Hazard Game, a controlled...

<https://arxiv.org/abs/2607.23982>

28 CONF

Why Public Service AI Governance Frameworks Risk Failing in the Age of General-Purpose AI: Lessons from Policing

ArXiv CS.AI 50%

arXiv:2607.25648v1 Announce Type: cross Abstract: Public services face growing pressure to adopt artificial intelligence (AI) to close the gap between rising demand and falling resources. That pressure has intensified with general-purpose AI (GPAI): AI built on large language...

<https://arxiv.org/abs/2607.25648>

29 CONF

MyMentorLLM: A psychotherapy GenAI environment with multimodal voice/text patients, trainees and experts for deliberate practice

ArXiv CS.AI 50%

arXiv:2607.25667v1 Announce Type: cross Abstract: Psychotherapists need repeated training and supervision by experts; however, scalability is problematic. Here we present MyMentorLLM, a multimodal voice- and text-based simulation environment for deliberate practice, used to...

<https://arxiv.org/abs/2607.25667>

30 CONF

DynaBridge: Dynamic Summary-Guided Cross-Task Multimodal Fusion for DASS-Structured Mental Health Assessment

ArXiv CS.AI 50%

arXiv:2607.25679v1 Announce Type: cross Abstract: Multimodal behavioral analysis offers a scalable approach to assessing depression, anxiety, and stress, yet generic fusion models often ignore the psychometric structure of questionnaire labels. In DASS-21, risk labels are...

<https://arxiv.org/abs/2607.25679>

31 CONF

Rashomon Alignment

ArXiv CS.AI 50%

arXiv:2607.25680v1 Announce Type: cross Abstract: We propose Rashomon Alignment (RA), a new measure to assess functional similarity between two models. Existing functional similarity measures are distributional, quantifying differences between outputs of models applied to...

<https://arxiv.org/abs/2607.25680>

32 CONF

From Deterministic to Generative Deep Learning for Urban Air Quality Reconstruction from Sparse Observations

ArXiv CS.AI 50%

arXiv:2607.25687v1 Announce Type: cross Abstract: Full-field reconstruction of air pollution is essential for evaluating pollution exposure and supporting public health decision-making. However, the complex interactions among pollutants, hard-to-predict weather patterns, and...

<https://arxiv.org/abs/2607.25687>

33 CONF

Tools Are Not Islands: Set-Level Tool Retrieval for LLM Agents via Query-Conditioned Hyperedge Prediction

ArXiv CS.AI 50%

arXiv:2607.25718v1 Announce Type: cross Abstract: Large language model (LLM) agents increasingly rely on invoking external tools to complete real-world tasks. Tool retrieval, which selects a small task-relevant subset from a library of thousands of tools before the agent acts,...

<https://arxiv.org/abs/2607.25718>

34 CONF

OmniQEC: discovering practical quantum error-correcting codes by an AI scientist

ArXiv CS.AI 50%

arXiv:2607.25865v1 Announce Type: cross Abstract: Quantum error correction (QEC) is indispensable for scalable fault-tolerant quantum computing. However, discovering QEC codes that remain effective is challenging, as logical performance depends on the interplay between code...

<https://arxiv.org/abs/2607.25865>

35 CONF

How Do LLMs Read Bug Reports? An Empirical Study of Attention in LLMs for Automated Program Repair

ArXiv CS.AI 50%

arXiv:2607.25873v1 Announce Type: cross Abstract: Large Language Model (LLM)-based Automated Program Repair systems are advancing rapidly, yet their performance remains inconsistent. Even when provided with the same contextual information, an LLM may generate a correct patch for...

<https://arxiv.org/abs/2607.25873>

36 CONF

Device Invariance using Domain Adaptation on Acoustic Scene Classification

ArXiv CS.AI 50%

arXiv:2607.25887v1 Announce Type: cross Abstract: This paper explores the effectiveness of domain adaptation techniques when using convolutional neural network (CNN)-based and transformer-based feature representations for acoustic scene classification. Two well-known domain...

<https://arxiv.org/abs/2607.25887>

37 CONF

Minimizing Targeted Activations: Input-Only Suppression of Evaluation-Awareness Latents in Large Language Models

ArXiv CS.AI 50%

arXiv:2607.25907v1 Announce Type: cross Abstract: Activation steering controls model behavior by editing internal activations at inference time. We study its input-side dual: optimizing a fluent prompt so that a chosen internal latent is driven toward zero, with no...

<https://arxiv.org/abs/2607.25907>

38 CONF

Evaluating VLMs for Autonomous Agent-Driven Geometry Clipping Detection in Video Game QA

ArXiv CS.AI 50%

arXiv:2607.25921v1 Announce Type: cross Abstract: In this work, we study the use of Vision-Language Models (VLMs) for anomaly detection in an agent-driven game Quality Assurance (QA) pipeline focusing on geometry clipping. In this evaluation, a custom exploration agent navigates...

<https://arxiv.org/abs/2607.25921>

39 CONF

Face De-Identification: A Domain-Centric Survey from Capture to Processing

ArXiv CS.AI 50%

arXiv:2607.25926v1 Announce Type: cross Abstract: Face de-identification (De-ID) aims to remove or conceal personally identifiable facial features in images or videos to prevent identity recognition while preserving utility for downstream tasks. With the rising emphasis on data...

<https://arxiv.org/abs/2607.25926>

40 CONF

Knowledge-Guided Multimodal Reasoning over Interacting Streams for Video-Level Ambivalence and Hesitancy Recognition

ArXiv CS.AI 50%

arXiv:2607.25961v1 Announce Type: cross Abstract: Ambivalence and hesitancy (A/H) are conflicting affective states that precede the delay or abandonment of health behaviour change. Recognition of A/H at the video level is difficult, since the signal arises from disagreement...

<https://arxiv.org/abs/2607.25961>

41 CONF

MemLens: A Value-Aware Memory Management System with Interactive Analytics for LLM-based Agents

ArXiv CS.AI 50%

arXiv:2607.25992v1 Announce Type: cross Abstract: Recently, memory management has become a key infrastructure for LLM-based agents, as it directly affects long-horizon reasoning, personalized responses, and knowledge reuse. However, existing LLM memory systems typically adopt a...

<https://arxiv.org/abs/2607.25992>

42 CONF

Does Runtime Topology Context Improve LLM-Generated Kubernetes Security Patches?

ArXiv CS.AI 50%

arXiv:2607.25995v1 Announce Type: cross Abstract: Kubernetes is central to the cloud-native ecosystem, orchestrating containerised workloads. Recent work suggests that large language models (LLMs) can automate cluster security remediation, generating configuration patches from...

<https://arxiv.org/abs/2607.25995>

43 CONF

Empirical Evaluation of Out-Of-Distribution Performance of Tabular Foundation Models

ArXiv CS.AI 50%

arXiv:2607.26000v1 Announce Type: cross Abstract: Tabular Foundation Models (TFMs) have emerged as novel approaches for tabular predictive tasks, demonstrating competitive predictive performance to ensemble tree-based models. Most TFMs are trained and evaluated on independent...

<https://arxiv.org/abs/2607.26000>

44 CONF

MDTransformer: A Hardware-Software Co-Design of Mode-Division Photonic Transformer Accelerator with Inverse-Designed Coherent Crossbar

ArXiv CS.AI 50%

arXiv:2607.26016v1 Announce Type: cross Abstract: Recently, photonic transformer accelerators (PTAs) have successfully achieved significant speedup and energy efficiency improvements over electronic accelerators for expediting Transformer inference. However, state-of-the-art...

<https://arxiv.org/abs/2607.26016>

45 CONF

Diffusion Model-based Parameter Estimation in Dynamic Power Systems

ArXiv CS.AI 50%

arXiv:2411.10431v3 Announce Type: replace Abstract: Parameter estimation, which represents a classical inverse problem, is often ill-posed as different parameter combinations can yield identical outputs. This non-uniqueness presents a critical barrier to accurate and unique...

<https://arxiv.org/abs/2411.10431>

46 CONF

Controllable LLM Reasoning via Sparse Autoencoder-Based Steering

ArXiv CS.AI 50%

arXiv:2601.03595v2 Announce Type: replace Abstract: Large Reasoning Models (LRMs) exhibit human-like cognitive reasoning strategies (eg backtracking, cross-verification) during the reasoning process, which improves their performance on complex tasks. Currently, reasoning...

<https://arxiv.org/abs/2601.03595>

47 CONF

JobMatchAI-An Intelligent Job Matching Platform Using Knowledge Graphs, Semantic Search and Explainable AI

ArXiv CS.AI 50%

arXiv:2603.14558v3 Announce Type: replace Abstract: Recruiters and job seekers rely on search systems to navigate labor markets, making candidate matching engines critical for hiring outcomes. Most systems act as keyword filters, failing to handle skill synonyms and nonlinear...

<https://arxiv.org/abs/2603.14558>

48 CONF

DSevolve: Enabling Real-Time Adaptive Scheduling on Dynamic Flexible Job Shop with LLM-Evolved Heuristic Portfolios

ArXiv CS.AI 50%

arXiv:2603.27628v2 Announce Type: replace Abstract: In dynamic flexible job shops, order arrivals, machine breakdowns, and processing-time deviations continually reshape the scheduling state and the priority trade-offs behind dispatching decisions. Dispatching rules are well...

<https://arxiv.org/abs/2603.27628>

49 CONF

The Possibility of Artificial Intelligence Becoming a Subject and the Alignment Problem

ArXiv CS.AI 50%

arXiv:2604.14990v2 Announce Type: replace Abstract: The prospect of Artificial General Intelligence (AGI) is increasingly driving institutional decisions, and alignment of AGI is a hard problem. The currently dominant AI alignment strategies like reinforcement learning with...

<https://arxiv.org/abs/2604.14990>

50 CONF

Why Does Grounding Hurt Medical VQA? Benchmarking, Diagnosis, and Fine-Tuning of Vision-Language Models

ArXiv CS.AI 50%

arXiv:2604.27720v2 Announce Type: replace Abstract: Vision-language models (VLMs) are increasingly applied to medical visual question answering (Med-VQA), yet whether they can \emph{localize} the evidence behind their answers---a prerequisite for clinical auditability---is...

<https://arxiv.org/abs/2604.27720>

51 CONF

EviBack: Search-Agent Reinforcement Learning via Evidence-Constrained Teacher Backoff

ArXiv CS.AI 50%

arXiv:2607.23955v2 Announce Type: replace Abstract: Reinforcement learning enables Agentic RAG systems to learn multi-turn search from verifiable outcome rewards, but all- zero rollout groups provide no comparative signal and may hide useful search behavior. We present EviBack,...

<https://arxiv.org/abs/2607.23955>

52 CONF

"We'll have to see how it works": An interview study to understand collaborative practices in interdisciplinary artificial intelligence and healthcare research

ArXiv CS.AI 50%

arXiv:2311.18424v3 Announce Type: replace-cross Abstract: Developing artificial intelligence (AI) algorithms for healthcare is a collaborative effort, bringing data scientists, clinicians, patients and other stakeholders together. By understanding AI as 'sociotechnical' where...

<https://arxiv.org/abs/2311.18424>

53 CONF

A context-adaptive policy framework for robust and reactive robotic manipulation via uncertainty-aware imitation learning

ArXiv CS.AI 50%

arXiv:2410.24035v2 Announce Type: replace-cross Abstract: Generating robust and reactive manipulation strategies that can adapt to changing context information is a challenging task in robotics. Over the years, Learning from Demonstration (LfD) has emerged as an intuitive and...

<https://arxiv.org/abs/2410.24035>

54 CONF

Annotation-Assisted Learning of Treatment Policies From Multimodal Electronic Health Records

ArXiv CS.AI 50%

arXiv:2507.20993v4 Announce Type: replace-cross Abstract: We study how to learn treatment policies from multimodal electronic health records (EHRs) that consist of tabular data and clinical text. These policies can help physicians make better treatment decisions and allocate...

<https://arxiv.org/abs/2507.20993>

55 CONF

Comparing RAG and GraphRAG for Page-Level Retrieval Question Answering on a Math Textbook

ArXiv CS.AI 50%

arXiv:2509.16780v3 Announce Type: replace-cross Abstract: Large language models (LLMs) show promise as educational aids but often lack alignment with specific course materials. We investigate Retrieval-Augmented Generation (RAG) and GraphRAG for page-level question answering on...

<https://arxiv.org/abs/2509.16780>

56 CONF

Picasso: Holistic Scene Reconstruction with Physics-Constrained Sampling

ArXiv CS.AI 50%

arXiv:2602.08058v4 Announce Type: replace-cross Abstract: In the presence of occlusions and measurement noise, geometrically accurate scene reconstructions -- which fit the sensor data -- can still be physically incorrect. For instance, when estimating the poses and shapes of...

<https://arxiv.org/abs/2602.08058>

57 CONF

Structured Scaling of AI Discovery Across Diverse Scientific Domains

ArXiv CS.AI 50%

arXiv:2604.19341v2 Announce Type: replace-cross Abstract: Scientific discovery often requires many cycles of proposing, testing, and refining candidate solutions. Language models can increasingly participate in these loops, but simply generating more attempts does not ensure...

<https://arxiv.org/abs/2604.19341>

58 CONF

RoboMME-Interference: Benchmarking Robot Memory Under Interference

ArXiv CS.AI 50%

arXiv:2606.22338v2 Announce Type: replace-cross Abstract: Robots deployed in realistic settings will accumulate experience across many sessions and tasks over their deployment. The robot's tasks may often require it to remember information from multiple sessions ago, making...

<https://arxiv.org/abs/2606.22338>

59 **CONF****Filling Before Advancing: Capability-Gap-Driven Post-Training for Scenario-Specialized Remote Sensing MLLMs****ArXiv CS.AI 50%**

arXiv:2607.22205v2 Announce Type: replace-cross Abstract: Remote sensing multimodal large language models (RS-MLLMs) have improved general aerial-image understanding. However, Earth observation applications require fine-grained scenario specialization, constrained by scarce...

<https://arxiv.org/abs/2607.22205>